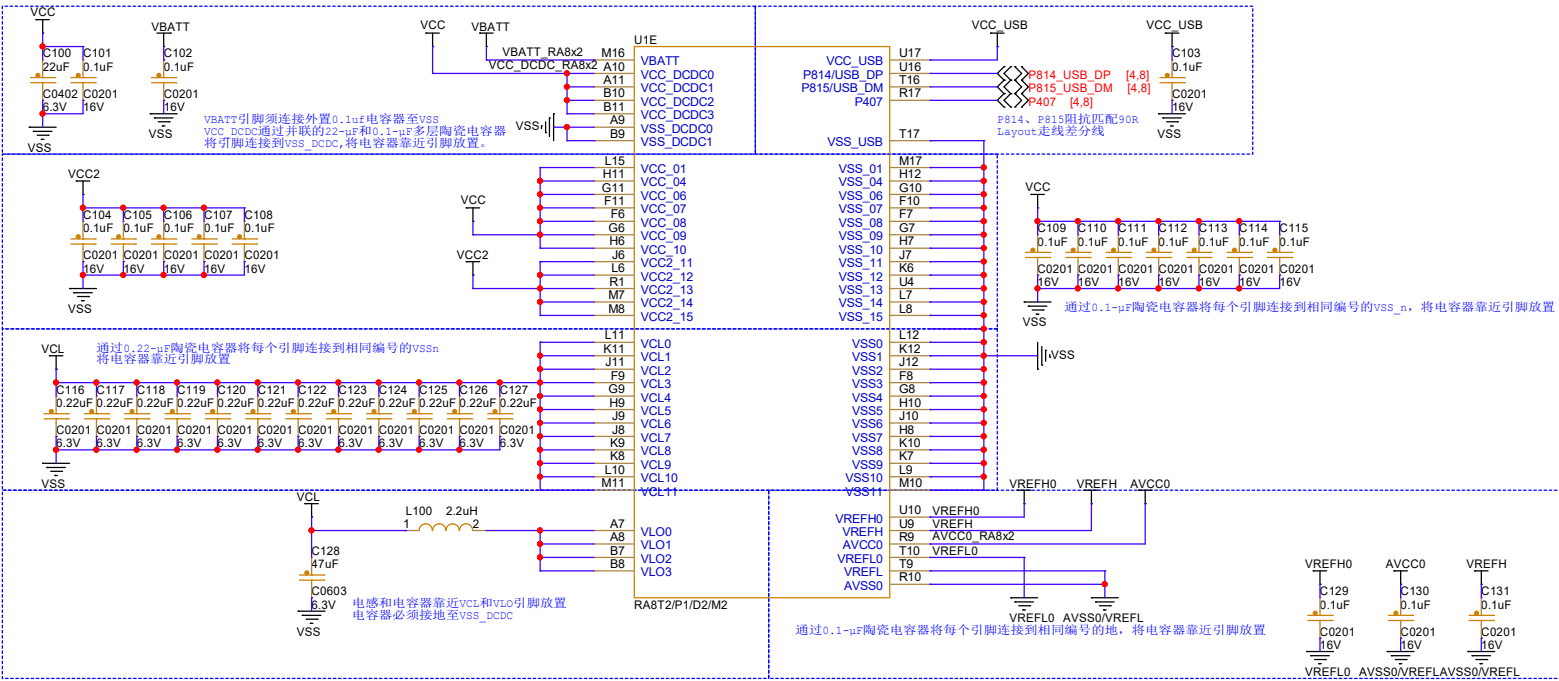




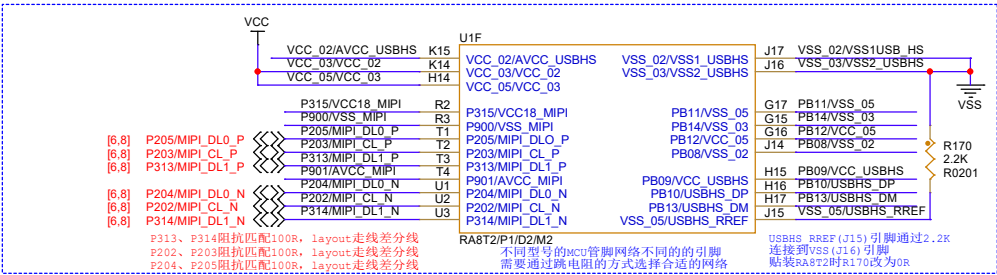
www.gitee.com/
ramcu/
cpk_examples



www.github.com/
renesas/
cpk_examples

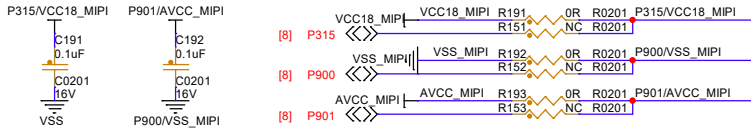


核心板芯片		
核心板芯片	R7KA8T2L	R7KA8P1K
C168-169	0.1uF	NC
C170	2.2uF	2.2uF
C171-C174	NC	0.1uF
C191-C192	NC	0.1uF
R145-R150	NC	NC
R151-R168	0R	NC
R170	0R	2.20K 1%
L171	NC	120R 100MHz
R172-R180	NC	0R
R191-R193	NC	0R



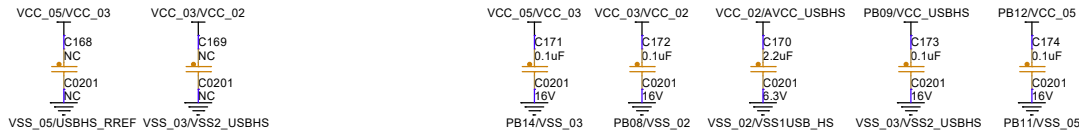
MIPI部分

连接电源的器件靠近MCU摆放，连接信号的器件在MCU-BTB间任意摆放

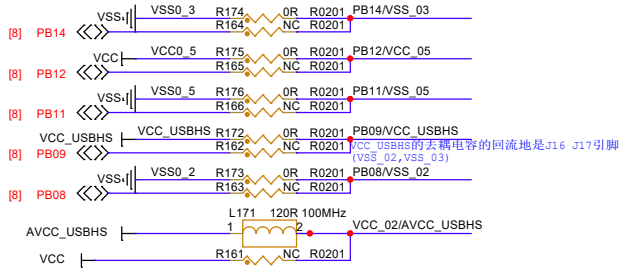


电阻选装

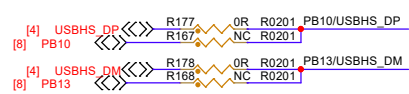
由于RA8P1/D2/M2和RA8T2的部分引脚网络不同，通过选装阻容的方式，应对不同型号的IC贴装。默认贴装RA8P1/D2/M2
连接电源的器件靠近MCU摆放，连接信号的器件在MCU-BTB间任意摆放



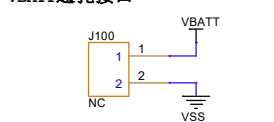
与电源相关的引脚，MCU信号出来后立即用电阻隔离



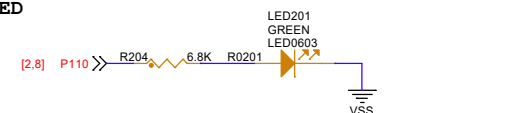
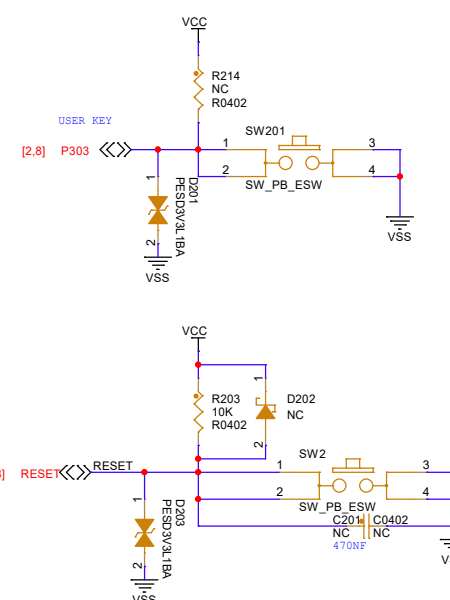
差分线走到J1(BTB)附近再用电阻隔离。



VBATT通孔接口



瑞萨电子www.renesas.com 东莞野火电子技术有限公司			
Title CPKCOR-RA8P1核心板			
Size A3	Document Number MCU 电源	Rev V1.1	
Date: Saturday, September 12, 2026	Sheet 1	of 10	

[illegible]

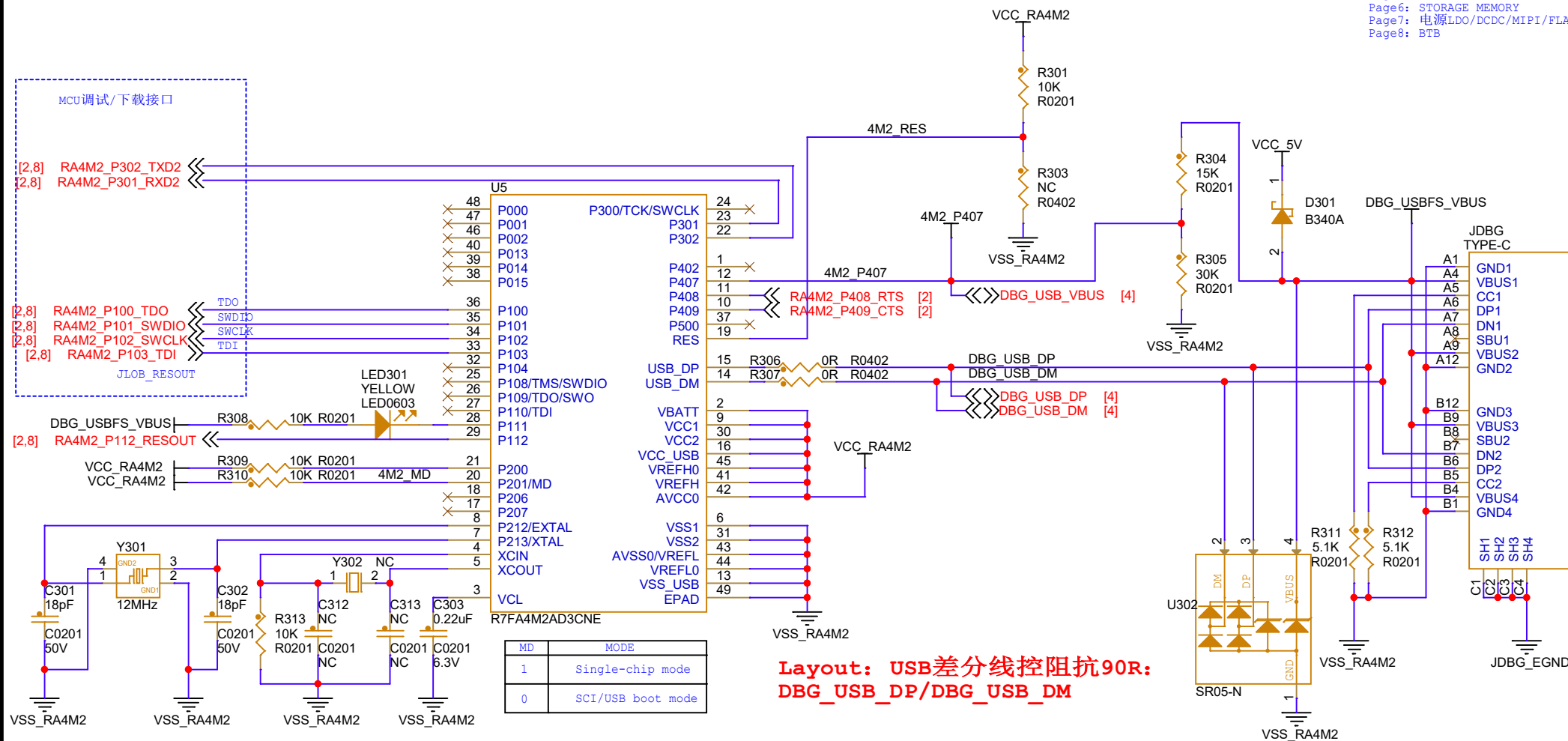
The schematic diagram illustrates the clock circuit for the PIC16F887. It consists of two inverters, Y201 and Y202, connected in a ring oscillator configuration. Y201 is a 24MHz oscillator with pins 4 and 1 connected to VSS, and pins 3 and 2 connected to Y202. Y202 is a 32.768kHz oscillator with pins 1 and 2 connected to VSS, and pins 3 and 4 connected to Y201. Both inverters have pull-up resistors (C204, C205, C206) to VSS. The output of Y202 is labeled XOUT and XIN.

如果使用USB HS，主晶振只能选12/20/24/48MHz

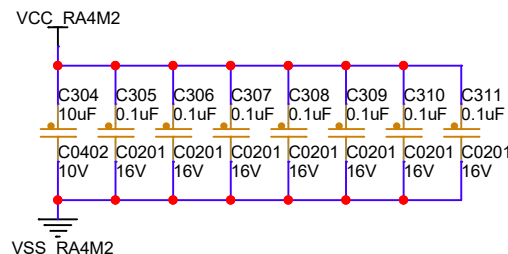
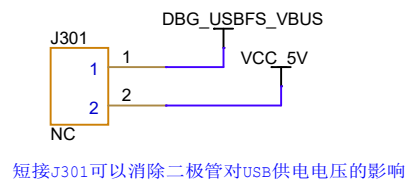
瑞萨电子 www.renesas.com 东莞野火电子技术有限公司			
Title CPKCOR-RA8P1核心板			
Size A3	Document Number	MCU IO	Rev V1.
Date:	Saturday, September 12, 2026	Sheet 2 of 10	

J-Link OB

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- Page4: USB HS
- Page5: SDRAM
- Page6: STORAGE MEMORY
- Page7: 电源LDO/DCDC/MIPI/FLASH
- Page8: BTB



注意：只能使用12MHz晶振

瑞萨电子 www.renesas.com

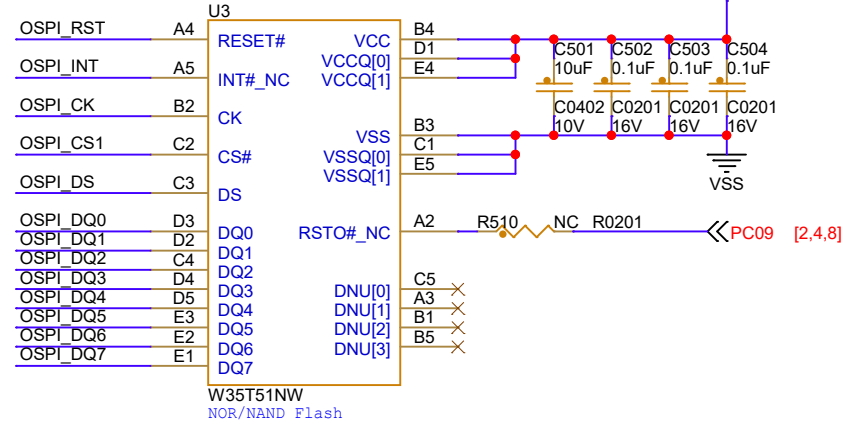
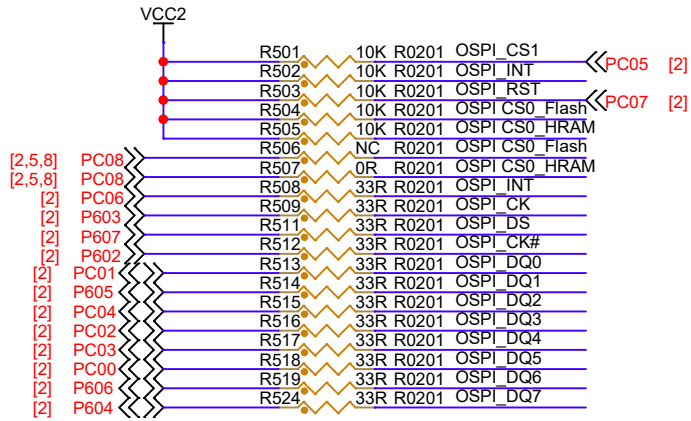
Title CPKCOR-RA8P1核心板			
Size A4	Document Number Jlink OB	Rev V1.1	
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如果使用USB HS，主晶振只能选12/20/24/48MHz



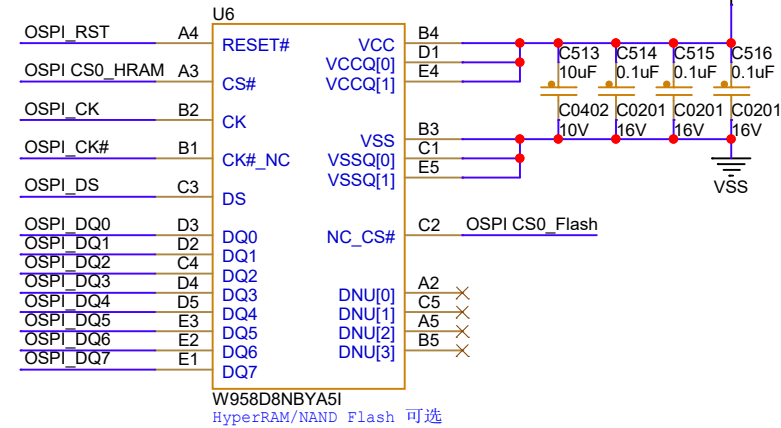
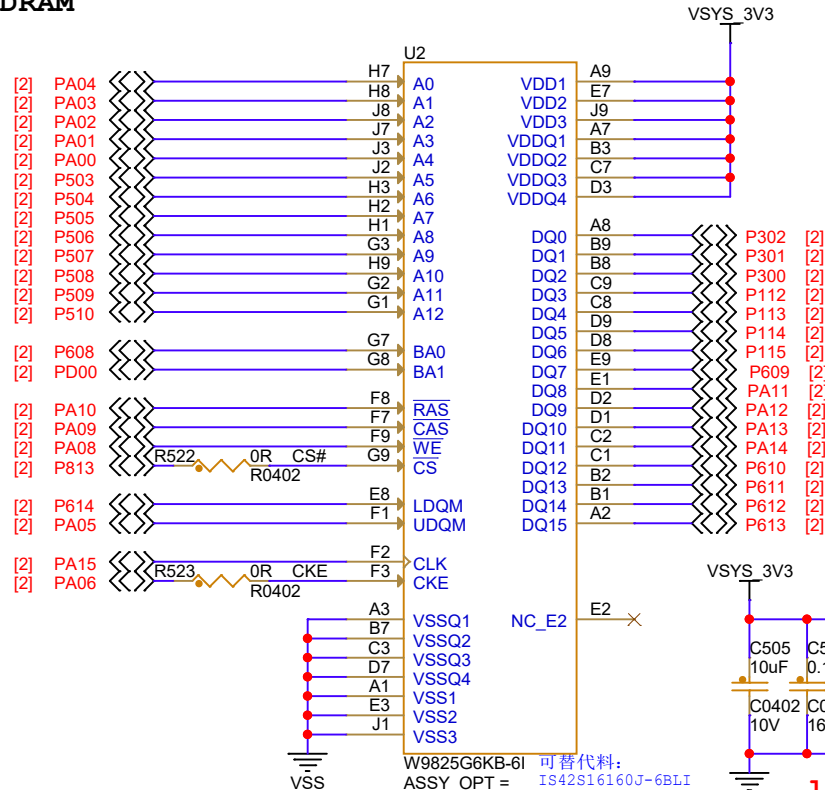
瑞萨电子 www.renesas.com 东莞野火电子技术有限公司			
Title CPKCOR-RA8P1 核心板			
Size A4	Document Number		Rev
	USB HS		V1.1
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FLASH



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Page5: SDRAM
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Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB

SDRAM

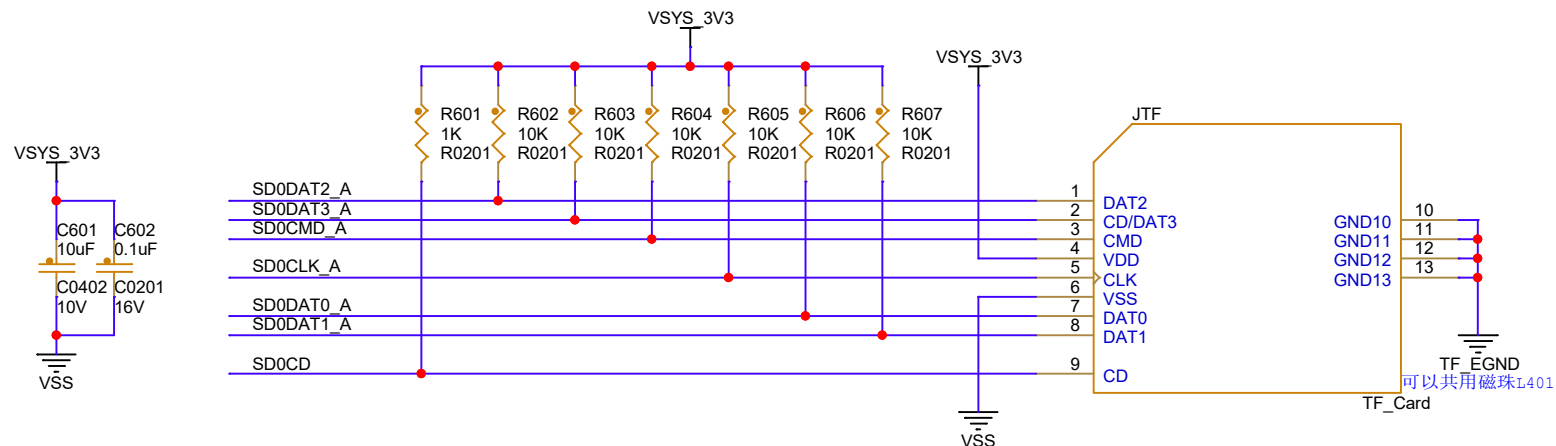


layout: SDRAM走线控阻抗50R

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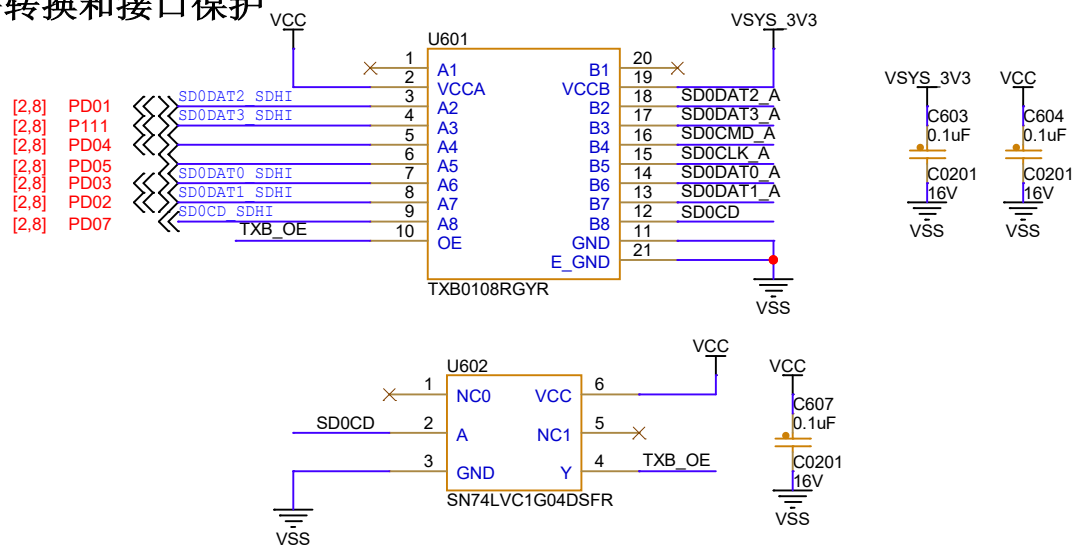
Title CPKCOR-RA8P1核心板		
Size A4	Document Number SDRAM/FLASH	Rev V1.1
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MicroSD卡槽

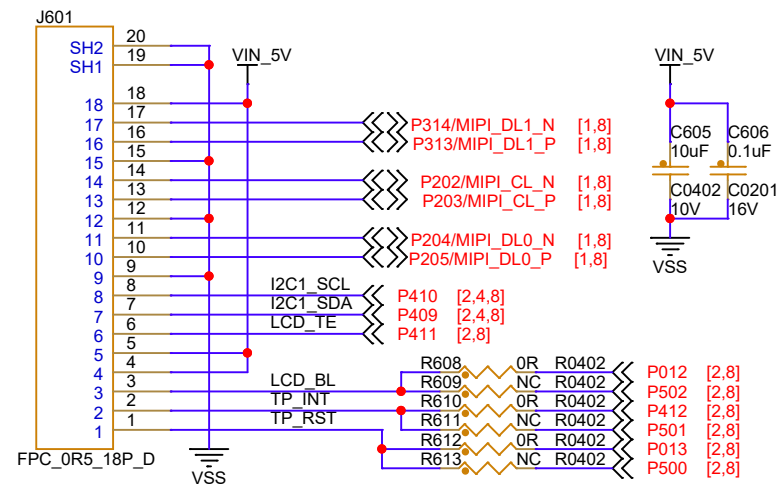


Page1: MCU 电源
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 Page7: 电源LDO/DCDC/MIPI/FLASH
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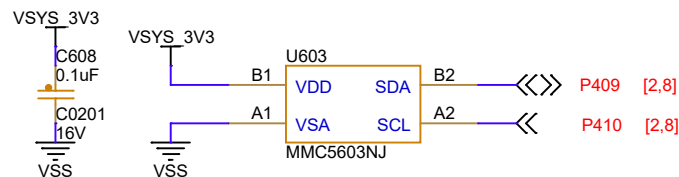
电平转换和接口保护



MIPI



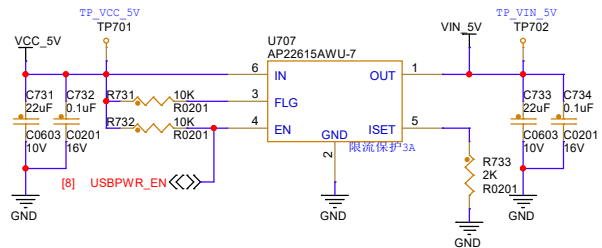
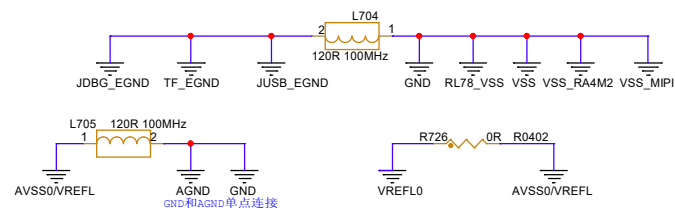
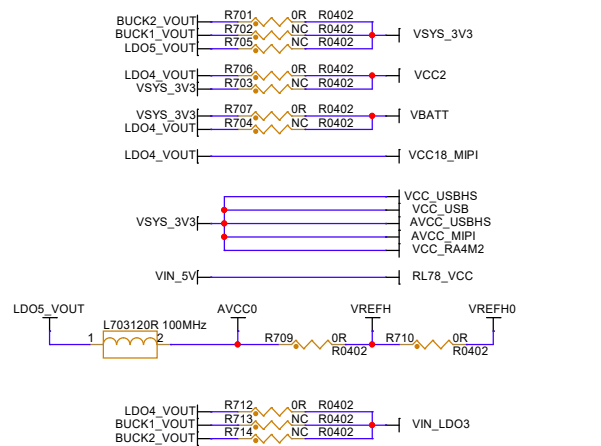
地磁传感器



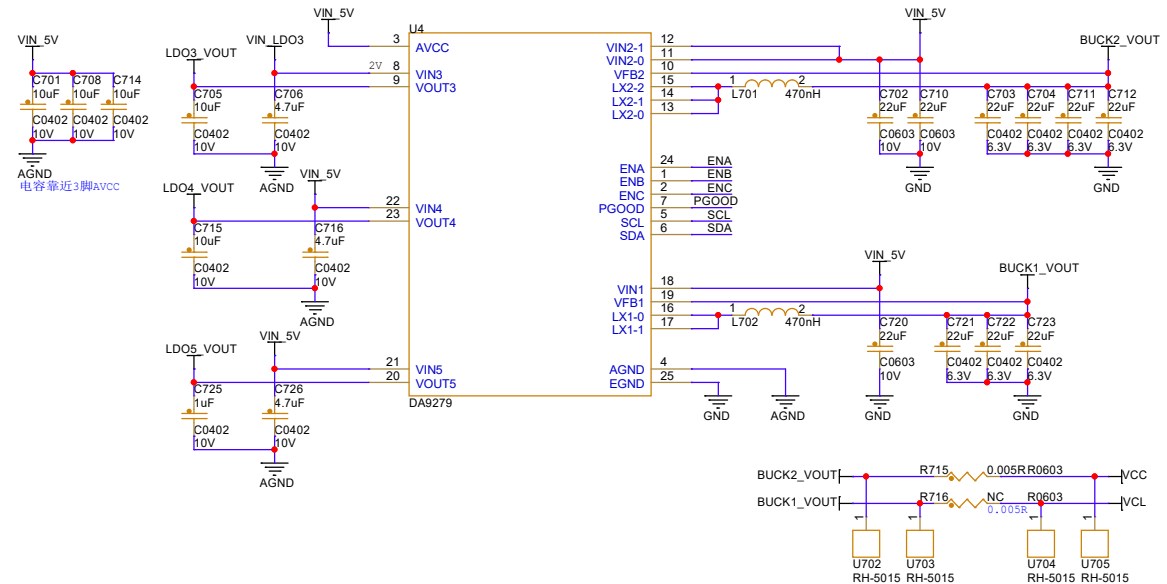
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Title CPKCOR-RA8P1核心板			
Size A4	Document Number SD CARD/MIPI		Rev V1.1
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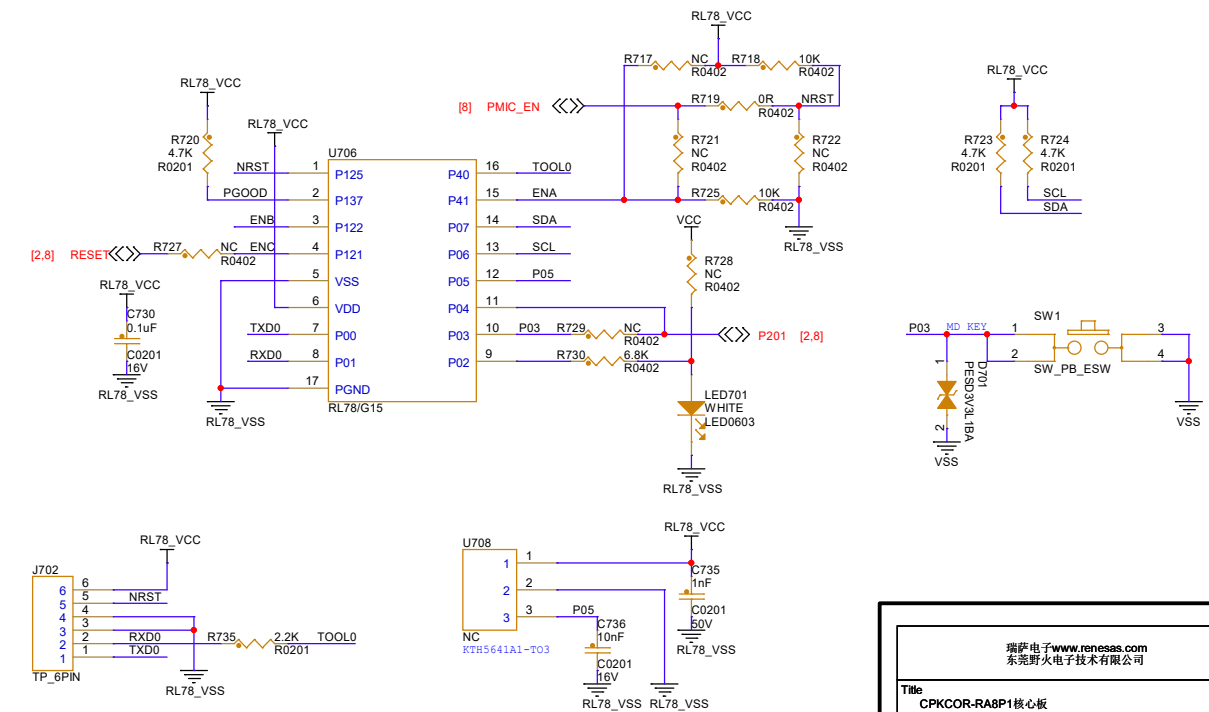
电源保护



PMIC

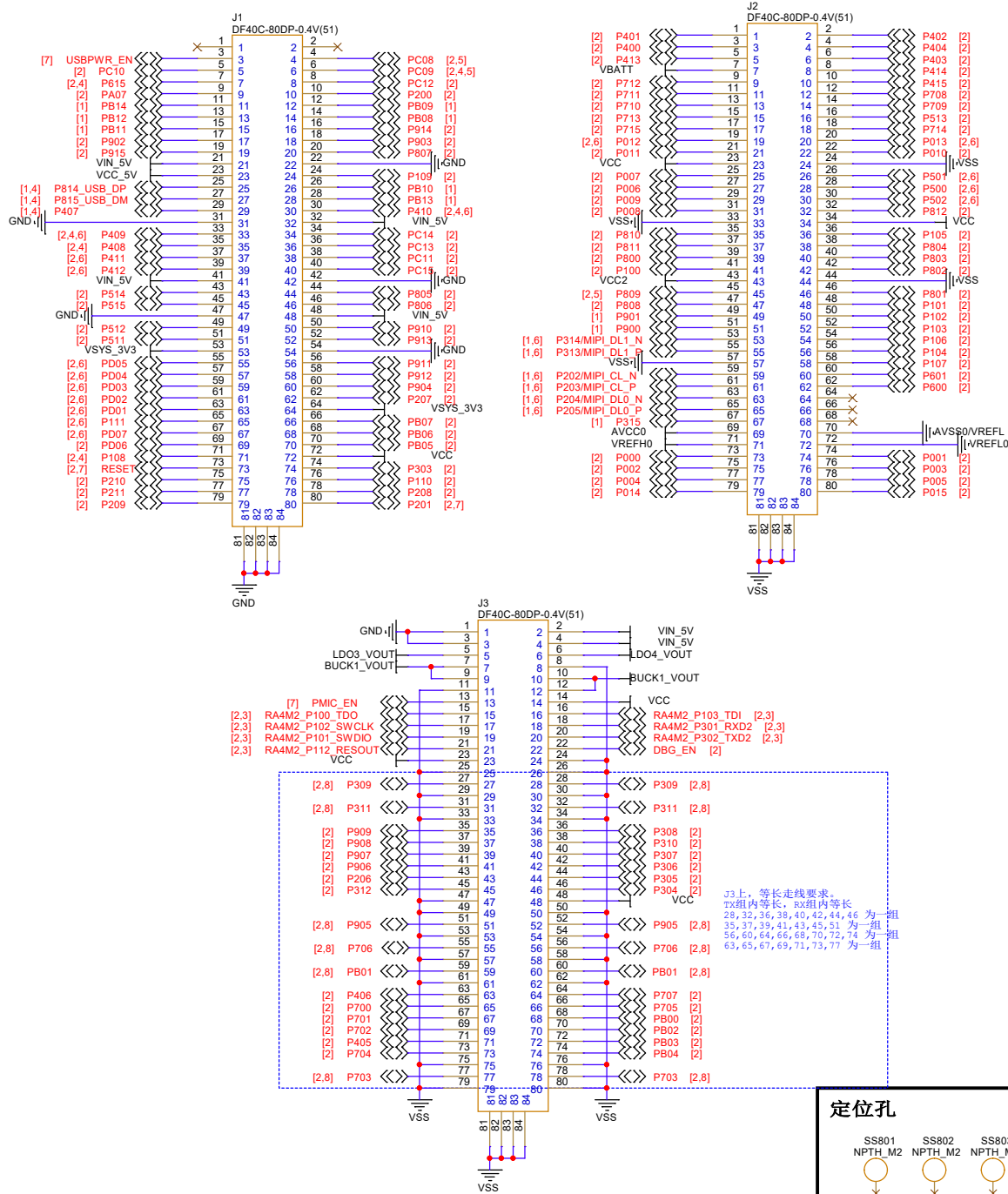


MCU



BTB座子

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Page8: BTB



历史版本

版本号	日期	设计	描述
V1.0	2026-06-08	llm	初版发行
V1.1	2026-09-04	llm	J1:34 修改为PC11, J1:38 修改为PC14

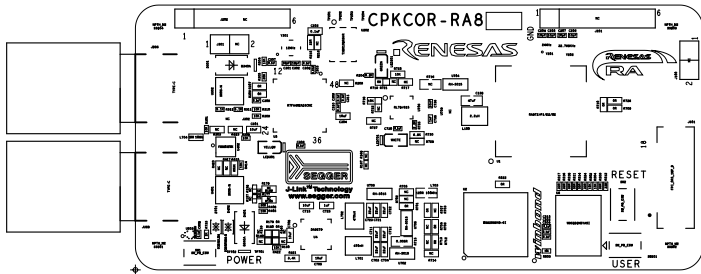
瑞萨电子www.renesas.com 东莞野火电子技术有限公司		
Title CPKCOR-RA8P1核心板		
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CPKCOR-RA8P1核心板_原理图_V1.1

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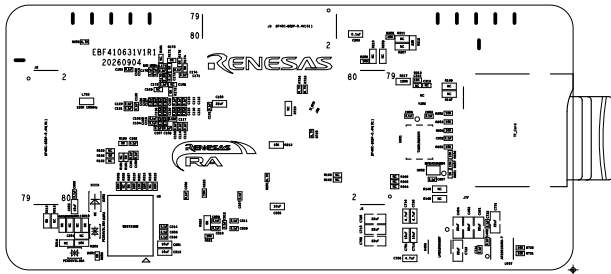
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ART FILM - ADT



ART FILM - ADT

ART FILM - ADB



ART FILM - ADB

CPKCOR-RA8P1 Power Supply Diagram

